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Quantum Physics

arXiv:2605.19117 (quant-ph)

[Submitted on 18 May 2026]

Quantum Magic Reveals CP Phases Invisible to Entanglement in Spin-0 Decays

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All standard scalar quantum-information measures -- concurrence, negativity, entanglement entropy, the optimized CHSH bound, and quantum Fisher information -- are CP-blind in ideal \\ spin-0 $\to f\bar{f}$ decays because the two-qubit spin state is maximally entangled for every CP angle. We show that stabilizer magic, fixed in the physical Pauli frame of spin analysis, escapes this blind spot: the stabilizer Rényi entropy admits an exact closed form, vanishing at CP-definite and Clifford phases and peaking at maximal non-Clifford mixing. Two experimentally accessible, magic-inspired CP witnesses follow; the linear amplitude is $14.3\times$ more efficient than its quartic counterpart and reaches discovery-level sensitivity at the HL-LHC for $H \to \tau^+\tau^-$.

Subjects: **Quantum Physics (quant-ph)**; High Energy Physics - Phenomenology (hep-ph); High Energy Physics - Theory (hep-th)

Cite as: **arXiv:2605.19117 [quant-ph]**

(or **arXiv:2605.19117v1 [quant-ph]** for this version)

https://doi.org/10.48550/arXiv.2605.19117

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From: Nicolas Viaux [view email]
[v1] Mon, 18 May 2026 21:07:10 UTC (96 KB)
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